cambI — White Paper (English)

Oct 17, 2026 · @Gustavo

Summary

cambI is a Brazilian real ↔ digital dollar exchange where customers trade with a verified vault, 24/7, see the full fee before confirming, and get a public receipt for every operation. Under the hood it is a pool on Solana; at the door, money moves in and out through Pix via a partner licensed by Brazil's Central Bank.

- **Who it is for:** freelancers and small businesses paid from abroad, tourists, people who save in dollars, and apps that need an FX engine (B2B).
- **What sets it apart:** the customer can own a piece of the exchange. Depositors in **Rende** receive part of the fee on every swap; **Baleias** ("whales", the liquidity providers) keep the larger share and take losses first.
- **Safety:** only wallets holding an identity credential can trade; an outflow circuit breaker; a withdrawal queue with advance notice; governance by a consortium of 10 providers that sign 7 of 10.
- **Status (Oct 17, 2026):** program v2.5 live on Solana devnet, applied through a 7-of-10 multisig with a 1-hour timelock. Ecosystem test passed 33 of 33 steps, with a 2.7 s median confirmation and ~0.000005 SOL network cost per transaction.

There is no speculative token: pool units represent reals and dollars 1:1, and returns come from real FX fees.

The problem

In traditional FX, customers pay a lot without seeing how much, wait days, and keep none of the profit.

Bottleneck	How the customer experiences it
Hidden spread	"Zero fee" with 3% to 6% built into the quote (market estimate)
Business hours	Friday afternoon means Monday; settlement in 1 to 2 days
Getting paid from abroad	Payment platforms charge a fee plus spread and hold funds for days

Bottleneck	How the customer experiences it
SWIFT	1 to 5 days; intermediary banks deduct fees without notice
Crypto P2P	Unknown counterparty, scam and Pix-reversal risk, a different price in every ad
Concentrated profit	The FX gain goes entirely to the intermediary

The timing helps: in October 2026 a major card network announced stablecoin payments across hundreds of millions of merchants. What is missing is a cheap, transparent bridge between the digital dollar and the Brazilian real.

The solution: peer-to-pool FX

The customer does not trade with a person: they swap with a vault on the blockchain, at the oracle price, and money leaves only when theirs arrives, in the same transaction.

- **Regulated at the door:** reals enter and leave via Pix through the licensed partner, which runs KYC and issues an on-chain credential to the wallet. Without a credential, the program refuses swaps and deposits.
- **A DEX on the inside:** the program executes the swap 24/7, with the fee computed and split in public code.
- **Private, not anonymous:** the blockchain shows only an address and a credential (level, expiry, issuer). The partner knows who is who and answers to the regulator.
- **QR payments:** pay any Pix QR code with digital dollars (the merchant receives reals) and get paid from abroad via Solana Pay.
- **Own the exchange:** part of every fee goes to whoever left money in the pool.

Token economics

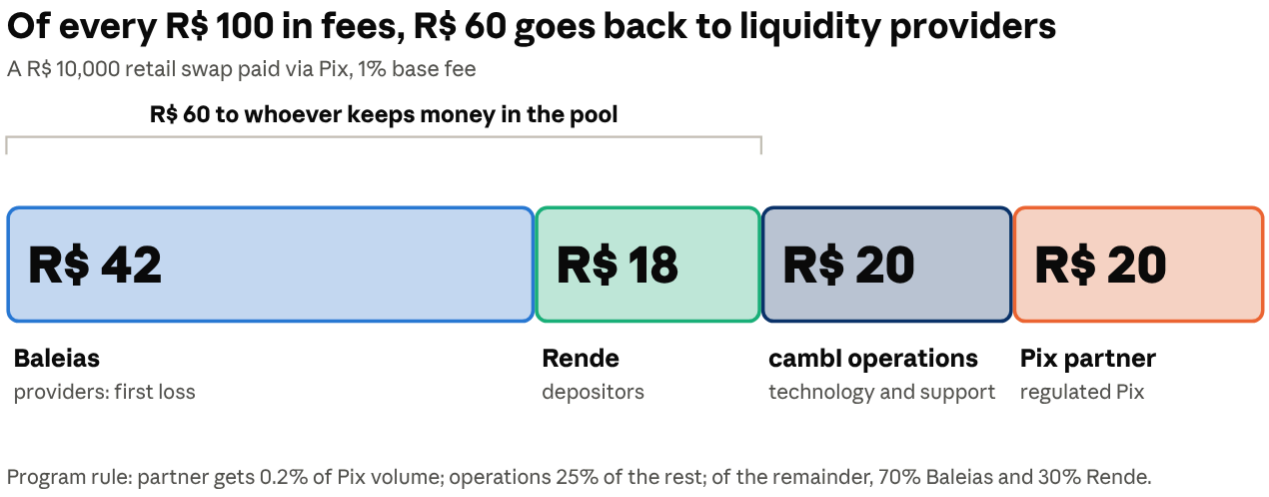
cambl has no speculative token and no governance token: the economy runs on real FX fees, split by a public rule between whoever operates, whoever handles the regulated edge, and whoever provides liquidity.

Units and backing

Unit	What it represents	On devnet	In production
Digital real	R$ 1	test cBRL, minted by the program	a real stablecoin from a regulated issuer, or tokenized real earning CDI
Digital dollar	US$ 1	test cUSD	USDC (or a tokenized dollar earning T-bill yield)
Share	A claim on one layer's fees, measured in reals at deposit	on-chain account per wallet	same

Principal is nominal in each currency: deposit US$ 100 and you are owed US$ 100. Shares only set your slice of the fees. With proof of reserves on, the pool stops accepting swaps if issued supply exceeds the attested reserve.

Where the fee goes



fee split rule of the cambi_pool program (default FeeConfig)

On devnet, the 13 test swaps generated R$ 39.40 in fees: R$ 17.02 to Baleias, R$ 8.10 to operations, R$ 7.29 to Rende and R$ 6.98 to the Pix partner. Partner apps (B2B) pay 0.4%, and depositors with R$ 1,000 or more pay 0.5%.

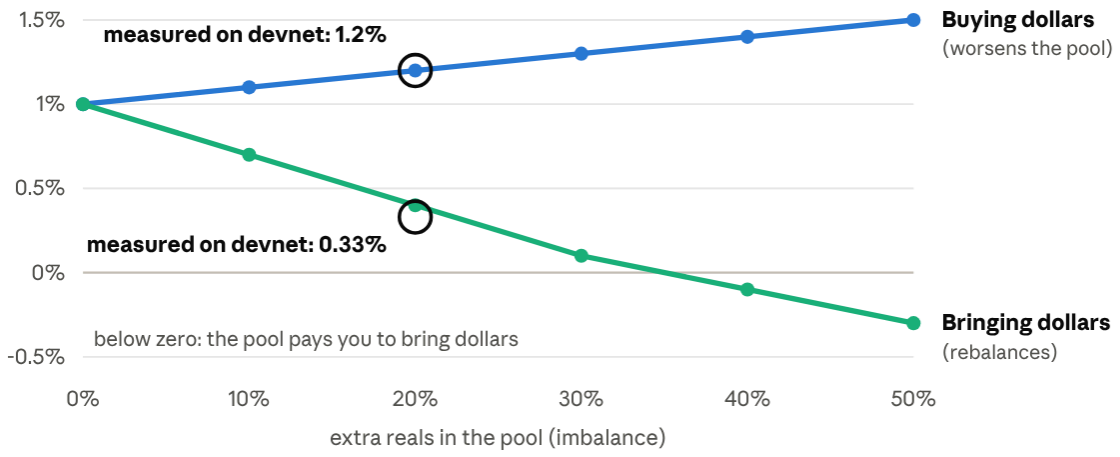
Dynamic fee, rebalancing discount and scarcity premium

Three mechanisms push the pool back toward balance, always in favor of whoever brings the scarce currency:

1. **Dynamic fee:** the base fee is multiplied by 1 + the imbalance the swap makes worse, between 0.7× and 1.5×.
2. **Rebalancing discount:** whoever brings the scarce currency gets a better price, 2% per 100% of remaining imbalance, capped at fee + base. A round trip never turns a profit (proven in tests).
3. **Scarcity premium on deposits (v2.5, live):** whoever deposits the scarce currency gets extra shares, 10% per 100% of remaining imbalance, at most 10%. In a pool with 20% extra reals, depositing dollars earns about 2% more. The premium flips to reals automatically when reals run short.

The pool charges whoever takes dollars and pays whoever brings them

Effective fee (%) by how many extra reals the pool holds, retail swap with a 1% base fee



program quote() formula + devnet ecosystem test, Oct 16, 2026

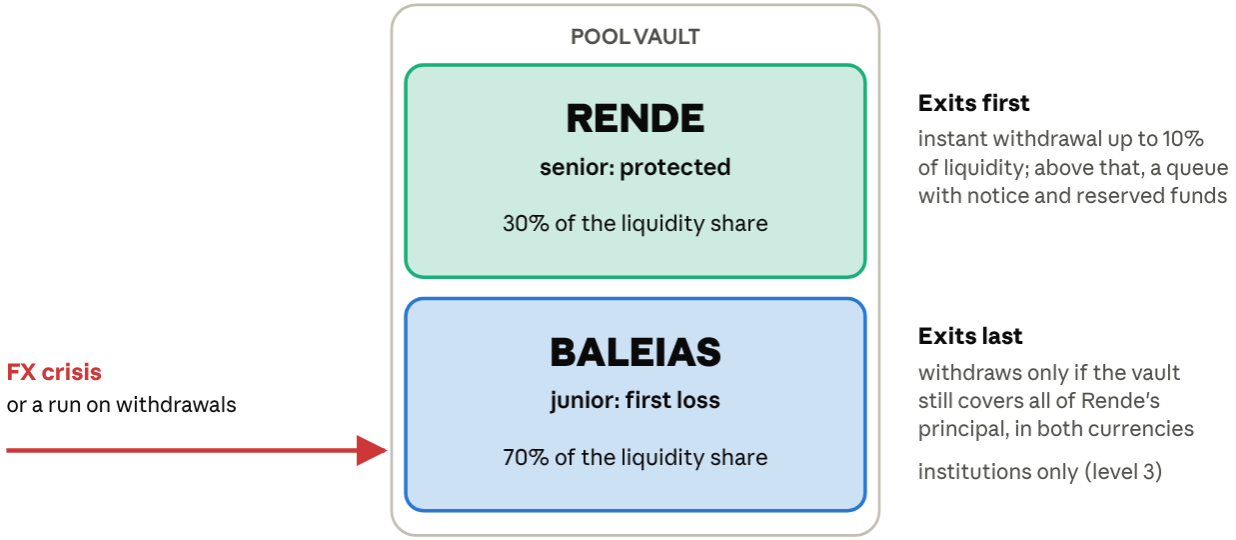
In the devnet test, the pool started with ~20% extra reals: buying dollars cost 1.2% and bringing dollars cost 0.33%. A balanced pool at launch and a retail base fee of 0.6% to 0.8% are the planned fixes.

Layers: Rende and Baleias

Rende is senior and Baleias are junior. Baleias keep 70% of the liquidity share because they take risk first: they can withdraw only if the vault still covers all of Rende's principal, in both currencies.

The Baleia takes the hit first; Rende stays protected

The pool vault in two layers: whoever earns more takes the risk first

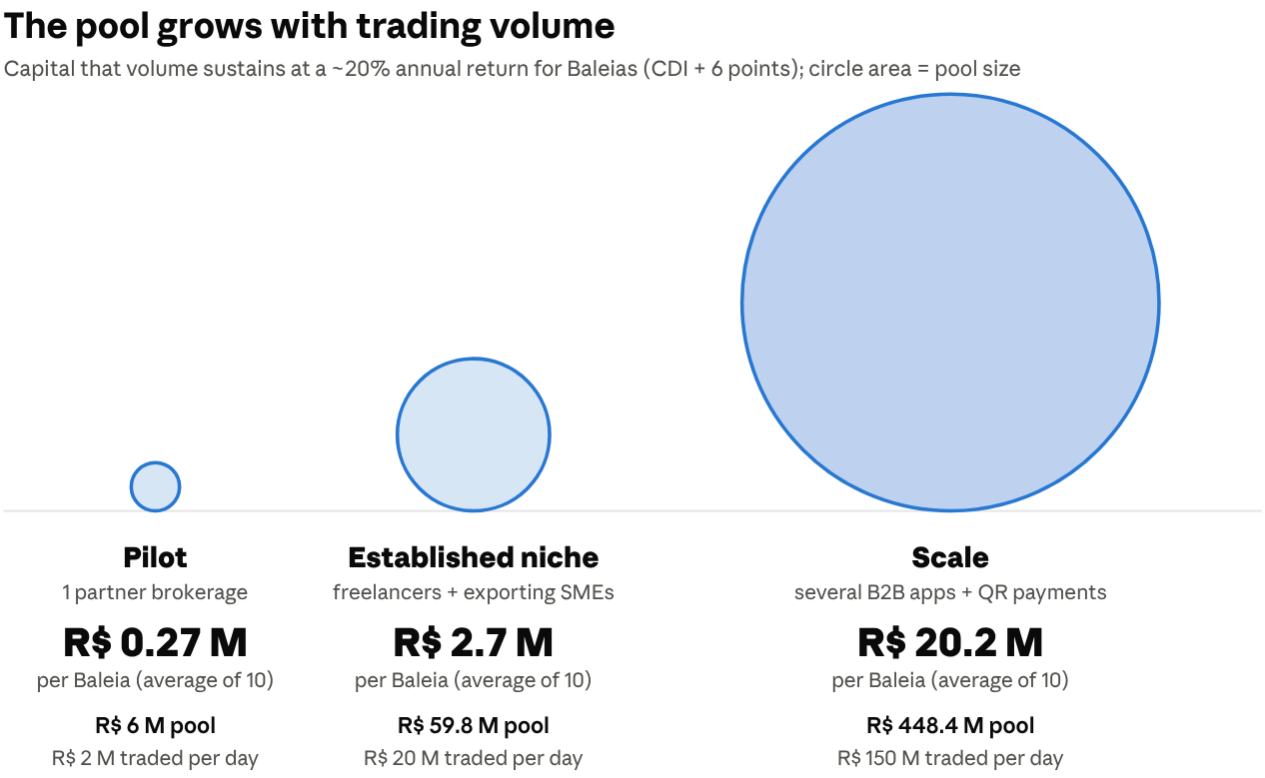


baleia_can_withdraw rule, withdrawal queue and level-3 credential in the program

The design resembles a fund with senior and subordinated tranches: those who want predictability sit in Rende; those who want higher returns and have their own FX flow sit in the Baleias.

Sizing and equilibrium

Baleia returns are fees divided by capital: more money than volume can sustain dilutes everyone. In the simulation, capital flows in until the return reaches CDI + 6 points (~20% a year), with fees compressed by competition.



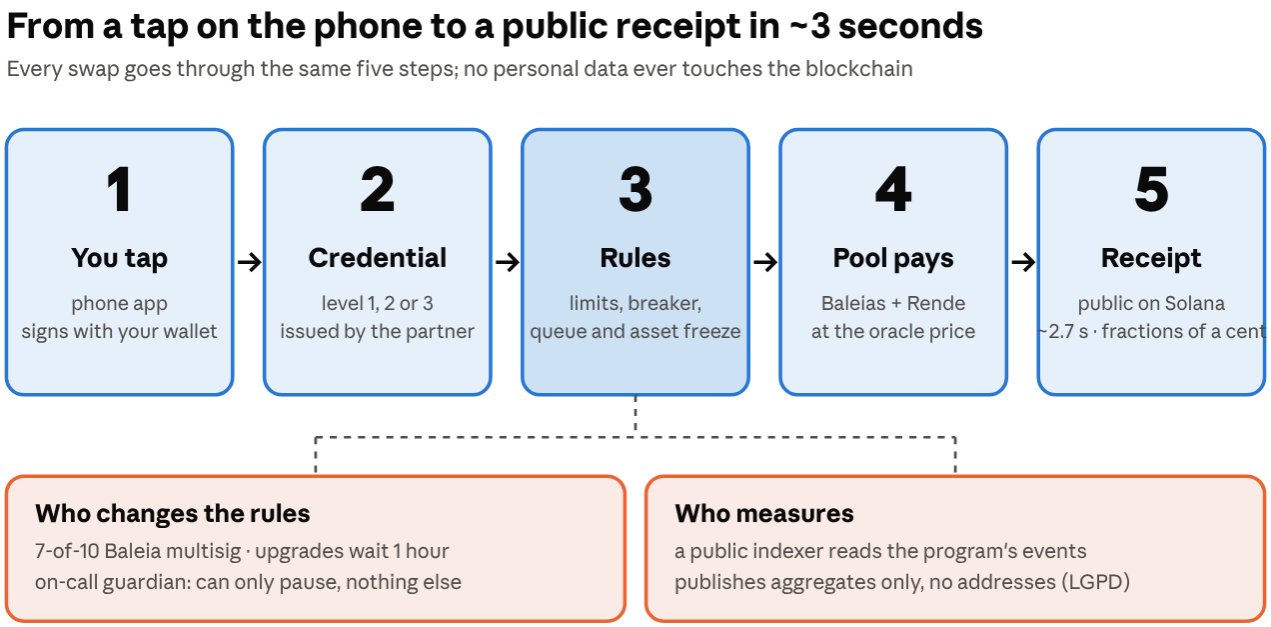
cambl simulation (baleias.py), fees compressed by competition, Oct 16, 2026

Capital beyond what volume sustains only dilutes. That is why the roadmap includes a capacity cap with an entry queue, and yield-bearing backing (tokenized Brazilian Treasury Selic bonds and US T-bills): idle money earns the base rate and FX fees become a bonus.

Simulation assumptions: 35% of volume retail, 55% B2B and 10% depositors; 252 business days; Baleias hold 45% of the pool; R$ 5.10 per dollar. These are hypotheses, not market data.

Architecture

cambl has four parts, and each does exactly one job.



cambi v2 architecture, v2 branch of the repository

App (React, mobile). Pix, QR, swaps, Rende and savings goals. It builds the program instructions, simulates before sending, and only then asks for a signature. It shows the reason for any refusal before the user taps the button.

Regulated partner. Runs KYC off-chain and issues the on-chain credential: level (1 basic, 2 full, 3 institutional), validity of up to 400 days, and issuer. The code↔person mapping stays with the partner only.

The `cambi_pool` program (Anchor, Solana). Holds the rules: valid credential, per-operation and 24-hour limits, a sliding-window outflow breaker, instant withdrawals up to 10% with a queue above that, asset freeze, proof of reserves, and the fee split. It is the only part that touches the vaults.

Governance and measurement. Rules change only through the Baleias' 7-of-10 multisig; code changes only through the upgrade multisig, with a 1-hour wait. A public indexer reads events and publishes aggregates only.

Component	Runs on	Devnet address
Program v2.5	Solana devnet	`3xdkEsUvcgRDKyqN8rY3RVVP6CRrtcuvLYz9iNE7rdhE`
Consortium pool	program account	`2jpkRKaQULxp6BMjuDxu3KhEBsaiDgmWbpe4JmiMkQbP`
Indexer and tests	GitHub Actions	`v2` branch

Security

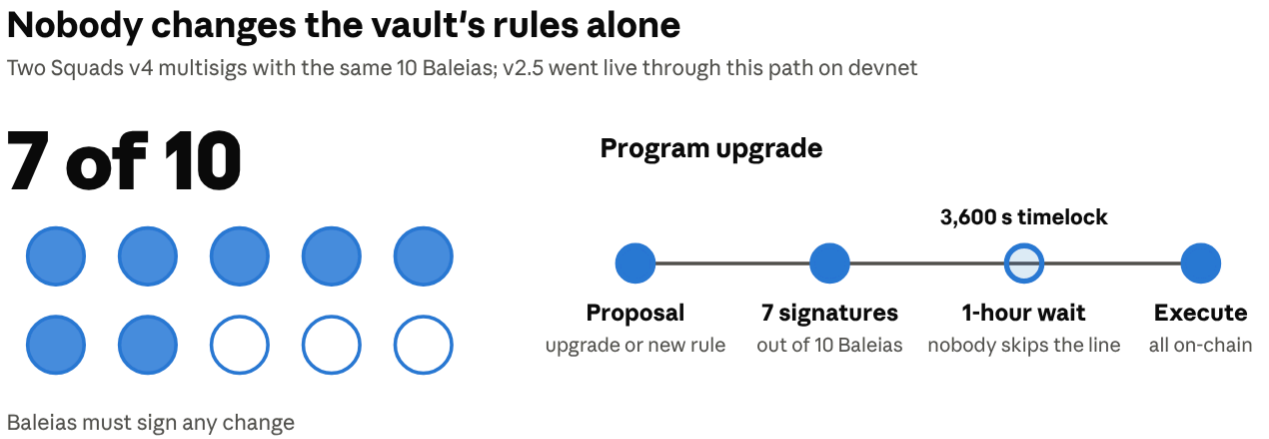
Every risk of on-chain FX has a safeguard written into the program and tested with real transactions.

Risk	Safeguard in the program	How it was tested
Unknown wallet trading	On-chain KYC credential issued by the partner; per-operation limit (R$ 10k level 1, R$ 100k level 2) and 24-hour limit (R$ 30k / R$ 300k)	Wallet without a credential refused on devnet (`KycRequired`)
Sanctions or court order	Asset freeze (Brazilian Law 13.810/2019): blocks swaps, deposits and withdrawals until released; revocation is separate	Integration test
Drain through a wrong price	Sliding-window breaker: at most 30% of each currency's liquidity leaves per hour, counting net outflow	Integration tests, including the round-trip attack
Bank run on the pool	Instant withdrawal up to 10% of liquidity; above that, a queue with advance notice and reserved funds	Integration test with a real wait
Leaked emergency key	The guardian can only pause; unpausing belongs to the 7-of-10 multisig	Devnet: guardian paused, tried to unpause and was refused; multisig unpaused in 10 s
Malicious code upgrade	Upgrade authority in a 7-of-10 multisig with a 1-hour timelock	v2.5 applied on devnet through this path; early execution refused
Unbacked currency	Proof of reserves: if issued supply exceeds the attested reserve, swaps and deposits stop	Integration test (off on devnet, no real custodian)
B2B flow out of control	Institutional per-operation limit and partner delisting	Integration test

An internal review of v2 found 8 issues, none critical, all fixed. An external audit is mandatory before real money.

Governance

The vault has no owner. It has a consortium.



Guardian: an on-call key that can only pause. Unpausing and changing price, fees or rules stay with the multisig.

Squads v4 multisigs on devnet; timelock checked by the upgrade script

- **Governance multisig (7 of 10).** The pool's admin: reference price, fees, limits, KYC issuer, unpausing, restoring credentials. Address `3LErideeXoJmJUQdXHeCXJ4dqFnLDPDrxb5sxu5EbdMu`.
- **Upgrade multisig (7 of 10, 1-hour wait).** The program's sole upgrade authority. At least 3,600 seconds pass between the seventh signature and execution, so anyone can see the change and withdraw first if they want. Address `2NyjSjF7iSbqqFfjCDxaTdsHK4GLDw99M1K5LMcSeEmp`.
- **Guardian.** An on-call key that can only pause swaps and deposits. It cannot unpause, touch prices or move money. Withdrawals keep working.

Proven on devnet: the v2.5 upgrade. On Oct 16, 2026, the scarcity premium went live through this path: proposal #2, 7 of 10 Baleia approvals, early execution refused by the timelock, execution after 61 minutes, and the on-chain binary checked against the compiled one.

The road there also produced a finding. Devnet refuses `ExtendProgramChecked` when called from inside a multisig (CPI), while `Upgrade` works fine. Since v2.5 is 3 KB larger, the extra program space is now added directly with the authority-free `ExtendProgram`, which only pays rent and changes no code, and the multisig proposal carries only the `Upgrade`. Governance is not weakened. We confirmed this with a disposable multisig and an empty test program before touching the real one.

Before mainnet: the 10 keys will belong to 10 different institutions (today they are test keys stored in the CI vault), and the guardian becomes a 24/7 alerting service with two people on call.

Compliance and regulation

The road to real money runs through a partner licensed by Brazil's Central Bank (BCB), with cambI as its technology. This is an informational reading, to be confirmed by a legal opinion.

Topic	Current reading	What cambI does
Virtual assets (Law 14.478/2022 and 2025 BCB rules)	Whoever operates the pool likely counts as a virtual asset service provider, needing prior authorization	Pilot as the technology of a licensed provider; admin keys under regulated governance
Foreign exchange (Law 14.286/2021)	Swapping a dollar-pegged stablecoin is an FX operation	Baleias are authorized institutions or qualified investors only (level-3 credential)
Securities (CVM Opinion 40)	Rende may be a public offering	In the pilot, Rende is for qualified investors only; no promised returns
Legal tender of the real	Obligations in Brazil are settled in reals	Paying a Pix QR with dollars: the merchant receives reals from the partner
AML/CFT and sanctions (BCB Circular 3.978, Law 13.810)	Monitoring, reporting to COAF, immediate asset freeze	Revocable credential, on-chain asset freeze, Travel Rule on transfers of US$ 1,000 or more
Data protection (LGPD, Constitutional Amendment 115/2022)	A wallet address is pseudonymous personal data	Nothing personal on-chain; the code↔person map stays with the partner
IOF tax	Rate on crypto operations still under debate	Configurable rate, shown separately from the fee

Three decisions come before launch: Rende's legal classification, backed by an opinion; the partner contract (who answers to the customer, IOF, COAF); and a formal data policy.

Measured results on devnet

Eight test users ran the consortium pool end to end: sign-up with a credential, swaps in both directions, Rende deposits and withdrawals, a guardian pause, and an unpause by the multisig.

Measured, not promised

Devnet ecosystem test, Oct 16, 2026; test tokens with no real value

33/33 **steps passed** 8 test users, live pool	**2.7 s** **confirmation (median)** p95 4.7 s; worst case 10 s	**0.000005** **SOL per transaction** a fraction of a cent
21.8k **compute units per swap** sign-up 64k; deposit 26k	**8** **weekly active wallets** the product north-star metric	**R$ 3,490** **swapped in 13 operations** R$ 39.40 in fees distributed

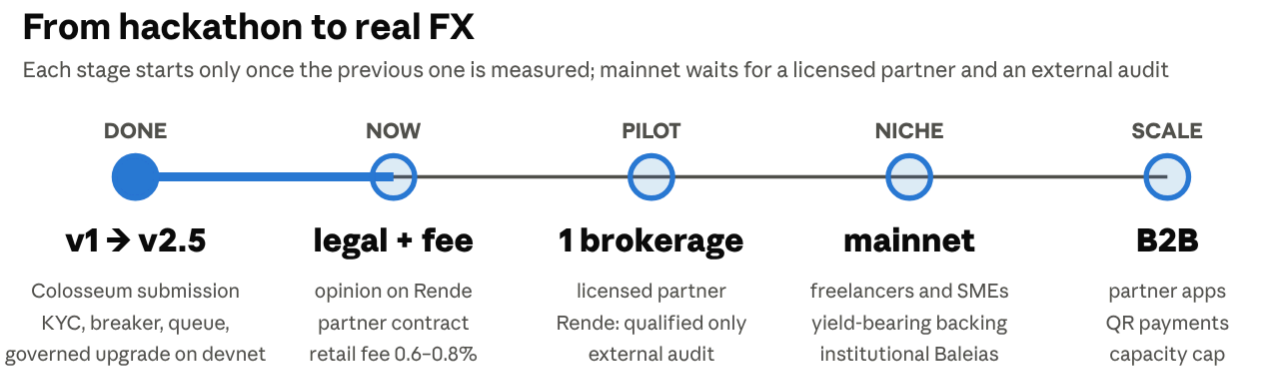
deployments/ecosystem-v2-report.json and deployments/metrics-v2.json, v2 branch, Oct 16, 2026

The finding that changed the product. Buying dollars cost **1.2%**; bringing dollars cost **0.33%.** The pool started with about 20% more reals than dollars, and the dynamic fee did its job: it charged whoever worsened the imbalance and rewarded whoever helped. That is too expensive for retail, which led to three fixes:

1. **Scarcity premium on deposits (v2.5, now live).** Depositing the scarce currency earns up to 10% more shares, in proportion to the imbalance, with no premium if the deposit overshoots.
2. **A balanced pool at launch,** seeded 50/50 by the Baleias.
3. **A retail base fee of 0.6% to 0.8%,** to be set by the multisig with pilot data.

The breaker ended the test using 2.2% of its limit in reals and 6.1% in dollars; no transaction was refused on-chain, because the app simulates before sending.

Roadmap



The program judged at Colosseum and the submission pool are unchanged: all of this lives on the v2 branch, in a new program.

cambl roadmap, October 2026

Stage	Delivery	Gate to move on
Done	v1 (submission); v2.4 and v2.5 on devnet: on-chain KYC, sliding breaker, withdrawal queue, guardian, multisigs, indexer, scarcity premium through a governed upgrade	33/33 ecosystem test; v2.5 executed after the 1-hour wait
Now	Legal opinion on Rende; model contract with a partner; retail base fee decided by the multisig	Opinion and contract signed
Pilot	One licensed brokerage as KYC issuer and Baleia; Rende for qualified investors only; external audit	Weekly active wallets growing; zero incidents
Niche	Mainnet with freelancers and exporting SMEs; yield-bearing backing (tokenized Selic bonds and T-bills)	Volume that sustains the Baleias
Scale	B2B apps, QR payments for tourists, capacity cap with an entry queue	Competitive retail fee without subsidies

Risks and disclaimer

Risk	What could happen	Mitigation
Oracle	A stale quote becomes profit for arbitrageurs and a loss for depositors (simulation: ~12% of the pool per year with hourly updates)	Deviation-triggered updates (~1 min), two sources agreeing within 0.3%, a cap on change per update
One-way flow	Brazilians buy dollars and the dollar side dries up	Dynamic fee, rebalancing discount, scarcity premium, Baleias restocking dollars
FX risk	The Baleia holds the open position between real and dollar	Baleias are junior and professional; an open-position cap on the roadmap
Smart contract	A bug in the code	Tests, internal review, external audit before real money, timelocked upgrades
Regulation	Rule changes, IOF tax, Rende's classification	Licensed partner, restricted Rende, configurable rates
Adoption	Low volume dilutes depositors' returns	Capacity cap, focus on B2B and inbound payments from abroad

Disclaimer. This document describes a prototype on a test network, using tokens with no real value. It is not an investment offer or a promise of returns. Returns depend on swap volume and can be negative for Baleias; deposits are not covered by Brazil's deposit guarantee fund (FGC). The legal and regulatory analysis here is informational and needs an opinion from licensed professionals. Market figures are estimates.